MAE 5036: Advanced Computational Solid Mechanics II

Problem set 2
Due OCT 23, Friday at 7:00 PM at Class

Please note: try to finish question 1 and 4 first. 2 and 3 can be done after the coming Wednesday lecture.

1. One-dimensional nonlinear heat conduction

Consider the nondimensional boundary-value problem

$$-\frac{d}{dx}\left[k(T)\frac{dT}{dx}\right] = 0, \qquad x \in (0,1), \qquad k(T) = 1 + 10T^2,$$

with $T(0) = 0$ and $T(1) = 1$. Use 10 uniform two-node linear elements and two-point Gauss quadrature. enforce the prescribed boundary temperatures throughout the iterations. Use a tolerance of 10^{-10} for the norm of residual vector. (Just use the code you developed in HW1)

(i) Derive the weak form, and Galerkin form, element residual vector, and consistent tangent matrix. (1PTs)

(ii) Solve using Newton–Raphson and modified Newton, by setting inital tempreture value to zero. Plot the temperature distribution and compare the residual histories. Does these methods both converge for this initial guess? (2PTs)

(iii) Solve using Newton–Raphson and modified Newton, by setting inital tempreture value as $T_0(x_i) = x_i$. Compare the iteration times and residual histories with part (ii). Does the choice of initial guess affect the convergence behavior? (2PTs)

2. Newton methods and line search

For parts (i) and (ii), consider the nondimensional internal force $N(d) = \arctan d$ and solve $N(d) = P$ for $P = 0.5$. Use an absolute equilibrium residual tolerance of 10^{-10} throughout this problem.

(i) Starting from $d_0 = 0.5$, compare the convergence of the Newton–Raphson and modified Newton methods. For modified Newton, hold the tangent fixed at $K(d_0)$. (2PTs)

(ii) Starting from $d_0 = 3$, a point where the tangent is soft, use Newton–Raphson without and with the line search, and compare your results. (2PTs)

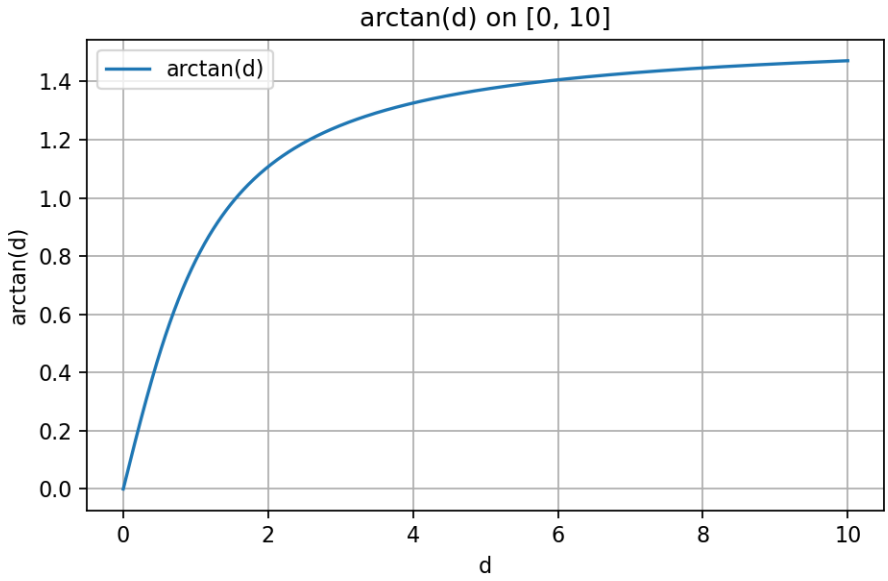


Figure 1: Nonlinear spring.

(iii) Consider a different the internal force

$$N(d) = \begin{cases} 100d, & d \le 0.1, \\ 10 + 10(d - 0.1), & 0.1 < d \le 0.3, \\ 12 + 100(d - 0.3), & d > 0.3. \end{cases}$$

Starting from $(d, P) = (0, 0)$, increase P to 20 in increments of $\Delta P = 1$. Use modified Newton without and with line search to obtain the converged force-displacement curves. Discuss your results. (1PTs)

3. The instability of the Von Mises truss: Newton–Raphson and arc-length control

When walking in the raining and windy day, one should be really careful about using umbrella. If the angle of it is slightly tilted, then the wind pressure can easily flip the umbrella inside out. Such phenomena illustrate the concept of instability, which is also observed in structural systems like the Von Mises truss. In this problem, we'll walk thourhg the steps to analyze the simple two-member truss structure and understand the snap-thourgh behavior of the equilibrium path (CLICK HERE for a demo).

Consider the symmetric two-bar truss in Fig. 2. The supports are fixed at $(-a, 0)$ and $(a, 0)$, and the apex is initially at $(0, H_0)$. A vertical downward force P acts at the apex. Restrict the response to symmetric deformation, so that the apex remains at $(0, H_0 - d)$, where d is positive downward. Both bars have constant axial rigidity EA. Use $a = 1$, $H_0 = 0.2$, and $EA = 100$ in consistent units.

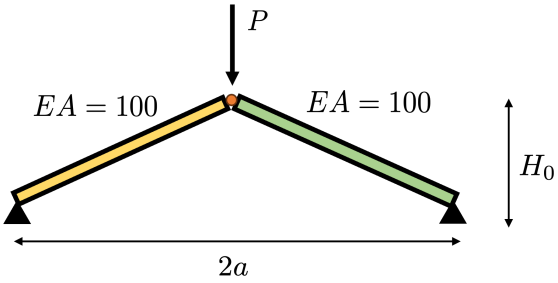




Figure 2: Von Mises truss.

Assume linear elastic material behavior and small axial strains in the bars, while retaining geometric nonlinearity due to finite rotations.

(i) **Internal force.** Derive the internal force resultant $N(d)$, where d is the vertical displacement of the apex. (1PT)

(ii) **Tangent stiffness.** Derive the corresponding tangent stiffness $K(d)$ and find out the expression for the critical tip displacement d_{cr} when the tangent stiffness is zero. Calculate the value based on the material and geometric properties. (1PT)

(iii) **Newton–Raphson method.** Starting from $(d, P) = (0, 0)$, prescribe increasing loads $P_n = n\Delta P$, with $\Delta P = 0.005$, up to $P = 1$. At each load step, solve $N(d) - P_n = 0$ using the standard Newton–Raphson method, updating the tangent stiffness at every iteration and using the previous converged displacement as the initial guess. Use $|N(d) - P_n| \leq 10^{-10}$ as the tolerance and a maximum of 50 iterations per step; record any failed steps. Plot the converged force–displacement points and report the iteration counts and residual histories near the limit point. Discuss any convergence failure or jump to a different equilibrium branch, and explain why increasing-load control cannot continuously trace the descending branch through the limit point.(1PT)

(iv) **Arc-length control.** Starting from $(d, P) = (0, 0)$, treat both d and P as unknowns and trace the equilibrium path until $d \geq 0.5$. and impose the arc-length constraint as

$\Delta a = 0.04$.

Use the consistent tangent and arc-length control algorithm to obtain the force displacement curve. Use an equilibrium tolerance of 10^{-10} and an absolute arc-length constraint residual tolerance of 10^{-8}. Compare the critical displacement with the results of part (ii) and the analytical relation derived in part (i). Explain why arc-length control can follow these static equilibrium states.(2PT)

4. Two-dimensional heat conduction: triangular and quadrilateral elements

Now, we would like to have a two dimensional problem. For the heat conduction, there is no difference comparing with the 1d bar problem in the first homework. Here the only difference is that the shape function depends on both x and y coordinates, and for the temperature in each element, the summation could involve more nodes.

Consider a rectangular plate occupying $\Omega = (0, 4) \times (0, 2)$, a uniform heat source $s = 10$, and conductivity

$$\mathbf{D} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

The strong form and boundary conditions are

$$\begin{cases} -\nabla \cdot (\mathbf{D}\nabla T) = 10, & (x, y) \in \Omega, \\ T = 10, & x = 4, \quad 0 < y < 2, \\ q_n = -\mathbf{n} \cdot \mathbf{D}\nabla T = -100, & y = 2, \quad 0 < x < 4, \\ q_n = -\mathbf{n} \cdot \mathbf{D}\nabla T = 0, & x = 0, \quad 0 < y < 2, \\ q_n = -\mathbf{n} \cdot \mathbf{D}\nabla T = 0, & y = 0, \quad 0 < x < 4. \end{cases}$$

Here $\mathbf{n}$ is the outward unit normal and q_n is the outward normal heat flux. Label the corner nodes as $1 : (0, 0)$, $2 : (4, 0)$, $3 : (0, 2)$, and $4 : (4, 2)$.

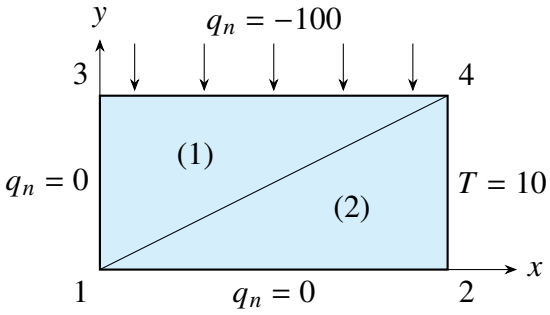


Figure 3: Plate geometry, boundary conditions, and two triangular elements.

(i) Derive the weak form using integration by parts and obtain the Galerkin finite element equations. Identify the element conductivity matrix and the load contributions from the heat source and prescribed boundary flux. (1PT)

(ii) Solve by hand using the two three-node linear triangular elements $(1, 4, 3)$ and $(1, 2, 4)$ shown in Fig. 3. Use the shape functions

$$N_i(x, y) = \frac{a_i + b_i x + c_i y}{2A_e}, \qquad i = 1, 2, 3,$$

where

$$a_i = x_j y_k - x_k y_j, \quad b_i = y_j - y_k, \quad c_i = x_k - x_j, \qquad 2A_e = \det \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix}.$$

Here (i, j, k) cycles through $(1, 2, 3)$, $(2, 3, 1)$, and $(3, 1, 2)$. The subscripts denote counterclockwise local node numbering; use local-to-global orders $(1, 4, 3)$ and $(1, 2, 4)$ for the two elements, respectively. Verify that these shape functions have the Kronecker-Delta property ($N_i(x_j, y_j) = 1$ when $i = j$, $\quad N_i(x_j, y_j) = 0$ when $i \neq j$). Compute the temperature at node 1 and 3. (2PTs)

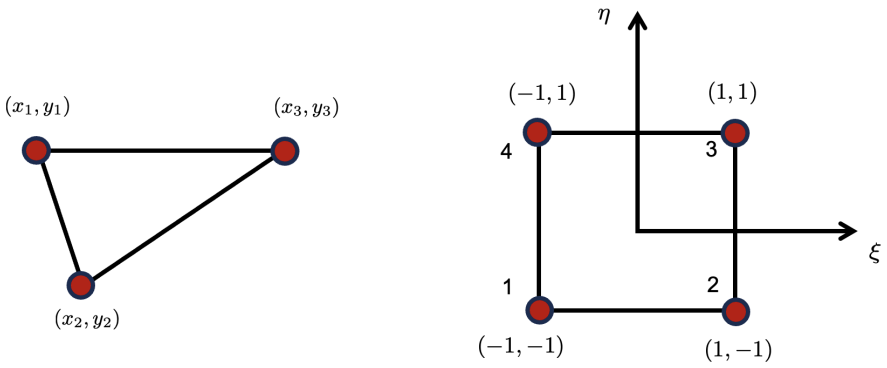


Figure 4: Triangular and quadrilateral elements.

(iii) Solve for the temperature by using the isoparametric approximation with one four-node bilinear quadrilateral element. For the local order shown in the picture, use

$$N_1 = \tfrac{1}{4}(1 - \xi)(1 - \eta), \quad N_2 = \tfrac{1}{4}(1 + \xi)(1 - \eta),$$
$$N_3 = \tfrac{1}{4}(1 + \xi)(1 + \eta), \quad N_4 = \tfrac{1}{4}(1 - \xi)(1 + \eta),$$

Compute $x^h = \sum_{A=1}^{4} N_A(\xi, \eta) x_A$ and $y^h = \sum_{A=1}^{4} N_A(\xi, \eta) y_A$. Find out the expressions for $\frac{\partial N}{\partial x}$ and $\frac{\partial N}{\partial y}$. Compute the temperature at node 1 and 3. Compare the results with part (ii). (2PTs)

Show your calculations. Programming is not required for this problem.